# Reuben Abraham T. Georgi

*Email: reuben.abraham97@gmail.com • Phone No: On request • Personal Page: https://ra-georgi.github.io/*

Robotics engineer with 3 years of experience and a Master's degree in Control and Autonomy. Experience and interests centre on classical and learning-based techniques for control, planning, and perception. Striving to develop safe and reliable autonomous systems for real-world deployment.

## EDUCATION

**Purdue University** West Lafayette, USA
Master of Science in Aeronautics and Astronautics (MSAA) May 2021

**Punjab Engineering College** Chandigarh, India
Bachelor of Technology (B.Tech) in Aerospace Engineering October 2019

## WORK EXPERIENCE/INTERNSHIPS

**Simelabs (Astek Group)** Kochi/Bangalore, India
*Robotics System Engineer* February 2022 - Present

Mobile Robot Application Demo

- Developed a web-based mobile robot application using ROS2 and Turtlebot3 as a technology demo for the robotics team
- Utilized Nav2 for SLAM & navigation, PyQt for creating a local GUI and OpenCV for handling vision tasks incorporated in the demo

R&D Automation Project (At **West Pharmaceutical Services**)

- Developed proof of concept projects for:
  - Pose estimation using an Intel RealSense depth camera and Open3D for autonomous pick-and-place using a Yumi collaborative robot
  - Ensuring collision-free trajectories during assembly line restarts using Moveit 2 and ROS 2 to reduce manual intervention and programming
  - Remote visualization of assembly operations in real-time using RobotStudio to communicate with robots and PLC
- Simulated and programmed ABB (Yumi, GoFa) and Omron (TM5) robots for process checking, layout & cycle time optimization and assembly operations

**AbbVie Inc.** Chicago, USA
*Research Analyst/ Summer Intern* July - September 2020

- Conducted CFD simulations in M-Star CFD and Ansys to characterize the hydrodynamics of USP dissolution apparatus with results published in the Journal of Pharmaceutical Sciences

**Helicopter Division, Hindustan Aeronautics Limited** Bangalore, India
*Intern* January - May 2018

- Internship Project: Mathematical modelling of helicopter blades to simulate response and predict performance of an isolated rotor

**SPACE lab, Khalifa University** Abu Dhabi, U.A.E
*Intern* May - July 2017

- Assembled and programmed TETRIX Ranger using MATLAB and Arduino for line following tasks and set up a Pumpkin CubeSat kit using MPLAB

## SELECTED PROJECTS

***UAV Simulation and Motion Planning Framework***

- Developing a modular and extensible Python tool for simulating quadcopter motion and experimenting with different motion planning and control techniques
- Currently implemented features:
  - Simulation: Set actuator limits, actuator delays, waypoints, wind disturbances (constant or Gaussian), obstacles (spherical)
  - Planning: Dijkstra, A*, RRT*, iLQR (Iterative Linear Quadratic Regulator)
  - Control: Cascade PID, LQR (using JAX for autodiff), linear MPC (using CasADi for optimization)
  - Visualization: Plots of state variable evolution with time, animation of quadcopter flight

***Autonomous Mobile Manipulator Simulation***

- Designed and implemented a ROS 2 mobile manipulator system integrating Nav2, MoveIt 2, and Gazebo to enable autonomous navigation and manipulation
- Utilized BehaviorTree.CPP to react to dynamically spawned objects and orchestrate task-level actions

***3D SLAM***

- Implemented the extended and unscented Kalman filter (EKF, UKF) in Python for simultaneous localization and mapping in 3 dimensions
- Simulated scenarios with known correspondences and unknown number of landmarks and introduced uncertainty by adding noise to odometry and sensor measurements

***Quadcopter Design using Genetic Algorithms***

- Formulated a multi-objective problem to minimize cost and weight of a quadcopter subject to performance and integration constraints
- Developed MATLAB app for users to provide data and to visualize results, including the Pareto frontier which displays Pareto optimal solutions.

## TECHNICAL SKILLS

Languages: **MATLAB**, **Python**, **C++**

Packages/Libraries: **Nav2 (ROS 2), Moveit2 (ROS 2), Gazebo, NumPy, Matplotlib, OpenCV, JAX, BehaviorTree.CPP**

Other software: **ABB RobotStudio/RAPID**, **RoboDK**, **Simulink, ANSYS, Arduino**

## PUBLICATIONS

Alexander M. Kubinski, Gayathri Shivkumar, Reuben A. Georgi, Susan George, James Reynolds, Ricardo D. Sosa, Tzuchi R. Ju, **Predictive Drug Release Modeling Across Dissolution Apparatuses I and II using Computational Fluid Dynamics,** Journal of Pharmaceutical Sciences, Volume 112, Issue 3, 2023, Pages 808-819, ISSN 0022-3549, https://doi.org/10.1016/j.xphs.2022.10.027.

Alexander M. Kubinski, Ricardo D. Sosa, Gayathri Shivkumar, Reuben Georgi, Susan George, Eric J. Murphy, Tzuchi R. Ju, **Predictive dissolution modeling across USP apparatuses I, II, and III**, Journal of Pharmaceutical Sciences, Volume 114, Issue 6, 2025, 103765, ISSN 0022-3549, https://doi.org/10.1016/j.xphs.2025.103765.